

Discrete Mathematics

Set Theory



Prove by induction that for $n > 0$ and $a \neq 1$

$$1 + a + a^2 + a^3 + \dots + a^n = \frac{1 - a^{n+1}}{1 - a}$$

Soln basis of induction
let $p(n) = 1 + a + a^2 + \dots + a^n = \frac{1 - a^{n+1}}{1 - a}$

(i) for $n=1 \Rightarrow p(1) = 1 + a = \frac{1 - a^2}{1 - a} = \frac{(1+a)(\cancel{1-a})}{\cancel{1-a}} = 1 + a$

It is true for $n=1$

Induction

(ii) for $n=k$ (that is, we assume that $p(k)$ is true)
$$p(k) = 1 + a + a^2 + \dots + a^k = \frac{1 - a^{k+1}}{1 - a}$$

(iii) for $n=k+1$, prove that $p(k+1)$ is true

w.k.t $p(k) = 1 + a + a^2 + \dots + a^k = \frac{1 - a^{k+1}}{1 - a}$

adding a^{k+1} both side of $p(k)$

$$\Rightarrow 1 + a + a^2 + \dots + a^k + a^{k+1} = \frac{1 - a^{k+1}}{1 - a} + a^{k+1}$$

conclusion

$p(k+1)$ is true whenever $p(k)$ is true

from (i), (ii) and (iii)

we conclude

$p(n)$ is true for $\forall n \geq 1$

$p(k+1)$

$$\begin{aligned} &= \frac{1 - a^{k+1}}{(1 - a)} + (1 - a)a^{k+1} \\ &= 1 - \cancel{a^{k+1}} + \cancel{a^{k+1}} - a^{k+2} \\ &= \frac{1 - a^{k+2}}{1 - a} \end{aligned}$$

Definition of Set

- A *set* may be viewed as any well-defined collection of objects, called the *elements* or *members* of the set.
- One usually uses capital letters, $A, B, X, Y, \dots$, to denote sets, and lowercase letters, $a, b, x, y, \dots$, to denote elements of sets.

Specifying Sets

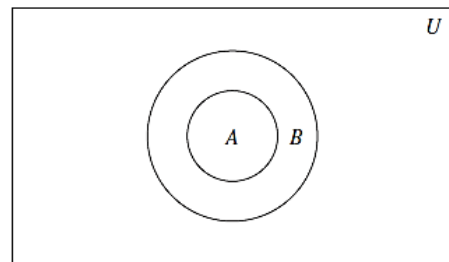
- There are essentially two ways to specify a particular set. One way, if possible, is to list its members separated by commas and contained in braces $\{ \}$. Example: $A = \{1, 3, 5, 7, 9\}$
- A second way is to state those properties which characterized the elements in the set. Example: $B = \{x \mid \underline{x \text{ is an even integer, } x > 0}\}$

B is the set of x such that x is an even integer and x is greater than 0

Subsets

The set A is a *subset* of B if and only if every element of A is also an element of B . We use the notation $A \subseteq B$ to indicate that A is a subset of the set B .

Example:



$$A = \{1, 3, 4, 7, 8, 9\}, \quad B = \{1, 2, 3, 4, 5\}, \quad C = \{1, 3\}.$$

Then $C \subseteq A$ and $C \subseteq B$ since 1 and 3, the elements of C , are also members of A and B . But $B \not\subseteq A$ since some of the elements of B , e.g., 2 and 5, do not belong to A . Similarly, $A \not\subseteq B$.

The Size of a Set

Let S be a set. If there are exactly n distinct elements in S where n is a nonnegative integer, we say that S is a *finite set* and that n is the *cardinality* of S . The cardinality of S is denoted by $|S|$.

Let A be the set of odd positive integers less than 10. Then $|A| = 5$.

$$A = \{1, 3, 5, 7, 9\}$$

Let S be the set of letters in the English alphabet. Then $|S| = 26$.

$$S = \{a, b, c, \dots, w, x, y, z\}$$

FINITE SETS, COUNTING PRINCIPLE

A set S is said to be finite if S is empty or if S contains exactly m elements where m is a positive integer; otherwise S is infinite.

Let E be the set of even positive integers, and let I be the *unit interval*, that is,

$$E = \{2, 4, 6, \dots\} \quad \text{and} \quad I = [0, 1] = \{x \mid 0 \leq x \leq 1\}$$

Then both E and I are infinite.

$$(0, 1)$$

$$\begin{array}{r} 0.1 \\ 0.01 \\ 0.001 \end{array}$$

Power Sets

Given a set S , the *power set* of S is the set of all subsets of the set S . The power set of S is denoted by $\mathcal{P}(S)$.

What is the power set of the set $\{0, 1, 2\}$?

Solution: The power set $\mathcal{P}(\{0, 1, 2\})$ is the set of all subsets of $\{0, 1, 2\}$. Hence,

$$\mathcal{P}(\{0, 1, 2\}) = \{\emptyset, \{0\}, \{1\}, \{2\}, \{0, 1\}, \{0, 2\}, \{1, 2\}, \{0, 1, 2\}\}.$$

Note that the empty set and the set itself are members of this set of subsets.

If a set has n elements, then its power set has 2^n elements.

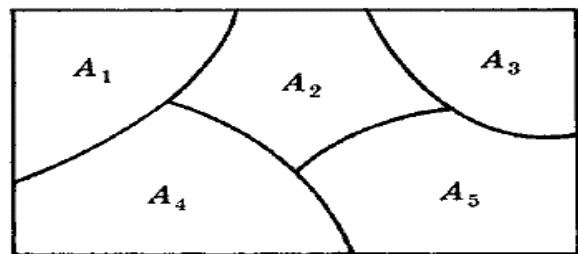
$$2^3 = 8$$

Partition of the Set

Let S be a nonempty set. A partition of S is a subdivision of S into nonoverlapping, nonempty subsets. Precisely, a partition of S is a collection $\{A_i\}$ of nonempty subsets of S such that:

- (i) Each a in S belongs to one of the A_i .
- (ii) The sets of $\{A_i\}$ are mutually disjoint; that is, if

$$A_j \neq A_k \quad \text{then} \quad \underline{A_j \cap A_k = \emptyset}$$



Consider the following collections of subsets of $S = \{1, 2, \dots, 8, 9\}$:

$$S = A_1 \cup A_2 \cup A_3 \cup A_4 \cup A_5$$

- (i) $\{\{1, 3, 5\}, \{2, 6\}, \{4, 8, 9\}\}$
- (ii) $\{\{1, 3, 5\}, \{2, 4, 6, 8\}, \{5, 7, 9\}\}$
- (iii) $\{\{1, 3, 5\}, \{2, 4, 6, 8\}, \{7, 9\}\}$

Then (i) is not a partition of S since 7 in S does not belong to any of the subsets. Furthermore, (ii) is not a partition of S since $\{1, 3, 5\}$ and $\{5, 7, 9\}$ are not disjoint. On the other hand, (iii) is a partition of S .

Set Operations

Let A and B be sets. The Cartesian product of A and B , denoted by $A \times B$, is the set of all ordered pairs (a, b) , where $a \in A$ and $b \in B$. Hence,

$$A \times B = \{(a, b) \mid a \in A \wedge b \in B\}.$$

What is the Cartesian product of $A = \{1, 2\}$ and $B = \{a, b, c\}$?

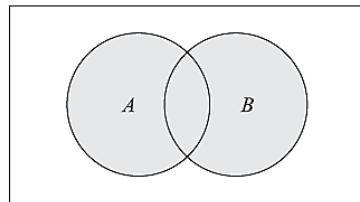
Solution: The Cartesian product $A \times B$ is

$$\underline{A \times B} = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}.$$

Union and Intersection

The *union* of two sets A and B , denoted by $A \cup B$, is the set of all elements which belong to A or to B ; that is,

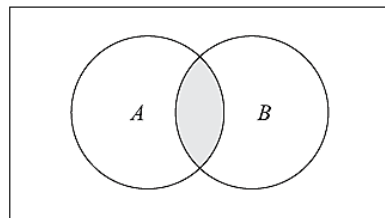
$$A \cup B = \{x \mid x \in A \text{ or } x \in B\} \quad \checkmark$$



(a) $A \cup B$ is shaded

The *intersection* of two sets A and B , denoted by $A \cap B$, is the set of elements which belong to both A and B ; that is,

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\} \quad \checkmark$$



(b) $A \cap B$ is shaded

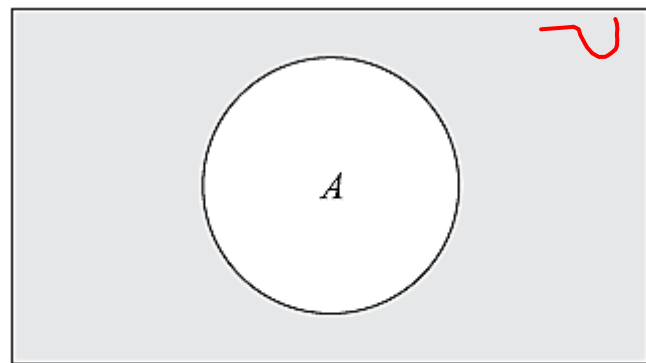
Let $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6, 7\}$, $C = \{2, 3, 8, 9\}$. Then

$$\begin{aligned} A \cup B &= \{1, 2, 3, 4, 5, 6, 7\}, & A \cup C &= \{1, 2, 3, 4, 8, 9\}, & B \cup C &= \{2, 3, 4, 5, 6, 7, 8, 9\}, \\ A \cap B &= \{3, 4\}, & A \cap C &= \{2, 3\}, & B \cap C &= \{3\}. \end{aligned}$$

complement or, simply, *complement* of a set A , denoted by A^C , is the set of elements which belong to U but which do not belong to A . That is,

$$A^C = \{x \mid x \in U, x \notin A\}$$

Some texts denote the complement of A by A' or $\bar{A}$.



Suppose $U = \mathbf{N} = \{1, 2, 3, \dots\}$ is the universal set. Let

$$A = \{1, 2, 3, 4\}, \quad B = \{3, 4, 5, 6, 7\}, \quad C = \{2, 3, 8, 9\}, \quad E = \{2, 4, 6, \dots\}$$

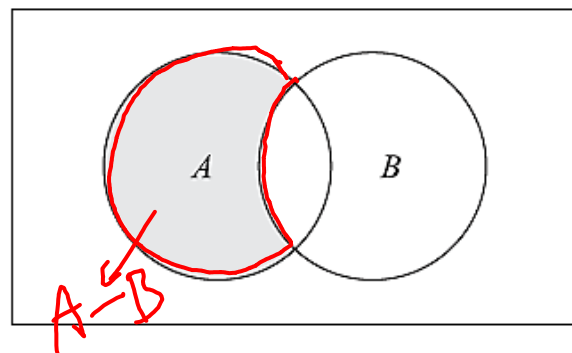
(Here E is the set of even integers.) Then:

$$A^C = \{5, 6, 7, \dots\}, \quad B^C = \{1, 2, 8, 9, 10, \dots\}, \quad E^C = \{1, 3, 5, 7, \dots\}$$

The *relative complement* of a set B with respect to a set A or, simply, the difference of A and B , denoted by $A \setminus B$, is the set of elements which belong to A but which do not belong to B ; that is

$$\underline{A \setminus B} = \{x \mid x \in A, x \notin B\}$$

The set $A \setminus B$ is read “ A minus B .” Many texts denote $A \setminus B$ by $A - B$ or $A \sim B$, which $A \setminus B$ is shaded.



(b) $A \setminus B$ is shaded

Suppose $U = N = \{1, 2, 3, \dots\}$ is the universal set. Let

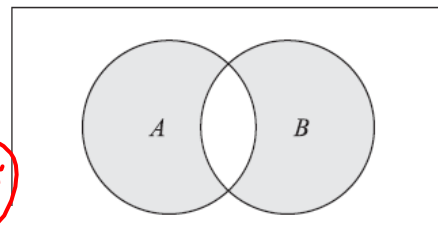
$$A = \{1, 2, 3, 4\}, \quad B = \{3, 4, 5, 6, 7\}, \quad C = \{2, 3, 8, 9\}, \quad E = \{2, 4, 6, \dots\}$$

$$A \setminus B = \{1, 2\}, \quad A \setminus C = \{1, 4\}, \quad B \setminus C = \{4, 5, 6, 7\}, \quad A \setminus E = \{1, 3\},$$

The symmetric difference of sets A and B , denoted by $A \oplus B$, consists of those elements which belong to A or B but not to both. That is,

$$A \oplus B = (A \cup B) \setminus (A \cap B) \quad \text{or} \quad A \oplus B = (A \setminus B) \cup (B \setminus A)$$

$A \Delta B$ $= (A - B) \cup (B - A)$



(c) $A \oplus B$ is shaded

Suppose $U = N = \{1, 2, 3, \dots\}$ is the universal set. Let

$$A = \{1, 2, 3, 4\}, \quad B = \{3, 4, 5, 6, 7\}, \quad C = \{2, 3, 8, 9\}, \quad E = \{2, 4, 6, \dots\}$$

$$\begin{aligned} A \setminus B &= \{1, 2\}, & A \setminus C &= \{1, 4\}, & B \setminus C &= \{4, 5, 6, 7\}, & A \setminus E &= \{1, 3\}, \\ B \setminus A &= \{5, 6, 7\}, & C \setminus A &= \{8, 9\}, & C \setminus B &= \{2, 8, 9\}, & E \setminus A &= \{6, 8, 10, 12, \dots\}. \end{aligned}$$

Furthermore:

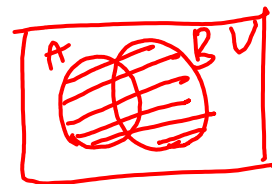
$$\begin{aligned} A \oplus B &= (A \setminus B) \cup (B \setminus A) = \{1, 2, 5, 6, 7\}, & B \oplus C &= \{2, 4, 5, 6, 7, 8, 9\}, \\ A \oplus C &= (A \setminus C) \cup (C \setminus A) = \{1, 4, 8, 9\}, & A \oplus E &= \{1, 3, 6, 8, 10, \dots\}. \end{aligned}$$

Inclusion-Exclusion Principle

Suppose A and B are finite sets. Then $A \cup B$ and $A \cap B$ are finite and

- $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
- That is, we find the number of elements in A or B (or both) by first adding $n(A)$ and $n(B)$ (inclusion) and then subtracting $n(A \cap B)$ (exclusion) since its elements were counted twice.

Note:



Suppose A, B, C are finite sets. Then $A \cup B \cup C$ is finite and

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$$



The generalized Inclusion-Exclusion principle

Let $A_1, A_2, \dots, A_n$ be n finite sets. Then

$$\left| \bigcup_{j=1}^n A_j \right| = \sum_{1 \leq i \leq n} |A_i| - \sum_{1 \leq i < j \leq n} |A_i \cap A_j| + \sum_{1 \leq i < j < k \leq n} |A_i \cap A_j \cap A_k| - \dots + (-1)^{n+1} \left| \bigcap_{j=1}^n A_j \right|.$$

A survey of 260 television viewers produced the following information: 64 watch sports, 58 watch films, 94 watch news, 28 watch both sports and films, 26 watch both sports and news items, 22 watch both films and news items and 14 are interested in all the three types. How many people in the survey are not interested in any of these three types? How many people in the survey are interested in seeing only news items?

Soln To find (i) number of people are not interested in any of these, three
(ii) number of people only interested in news

Given $n(U) = 260$

Let S denotes the number of people watch the Sports

Let F denotes the // Films

Let N denotes // News

$$n(S) = 64, n(F) = 58, n(N) = 94, n(S \cap F) = 28, n(S \cap N) = 26, n(F \cap N) = 22, n(S \cap F \cap N) = 14$$

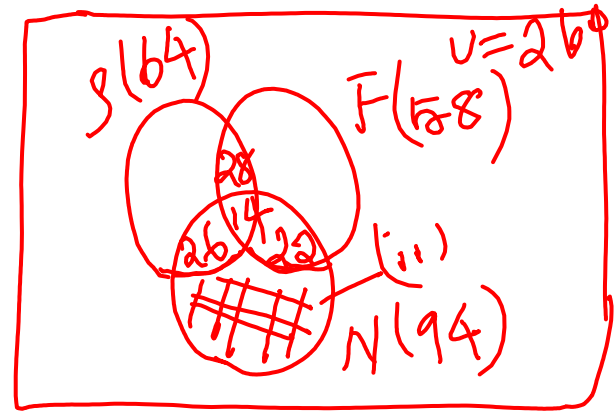
by inclusion and exclusion principle (no of people interested in any one)

$$n(S \cup F \cup N) = n(S) + n(F) + n(N) - n(S \cap F) - n(S \cap N) - n(F \cap N) + n(S \cap F \cap N)$$

$$= \cancel{58} + \cancel{64} + \cancel{94} =$$

$$= 58 + 64 + 94 - 28 - 26 - 22 + 14$$

$$= 154$$



(ii) no of people not interested in any three

$$= n(U) - n(S \cup F \cup N)$$

$$= 260 - 154 = 106$$

(ii) ✓ number of people watch only news

$$= 94 - 22 - 26 - 14$$
$$= 60$$

RELATIONS

Definition: Let A and B be sets. A binary relation or, simply, relation from A to B is a subset of $A \times B$.

- Suppose R is a relation from A to B . Then R is a set of ordered pairs where each first element comes from A and each second element comes from B .
- That is, for each pair $a \in A$ and $b \in B$, exactly one of the following is true:
 - (i) $(a, b) \in R$; we then say “ a is R -related to b ”, written aRb .
 - (ii) $(a, b) \notin R$; we then say “ a is not R -related to b ”, written aRb .

In fact, relationships can exist between any two sets, they are known as relations.

Consider the sets $\underline{A} = \{\text{Adithya, Priya, Thomas}\}$

and $\underline{B} = \{\text{IIT, NIT, IIIT}\}$.

Suppose Adithya is joined in IIT, Priya in NIT and Thomas in IIIT.

Let

$$\underline{R} = \{(\text{Adithya, IIT}), (\text{Priya, NIT}), (\text{Thomas, IIIT})\}$$

Using set builder form it can be defined as

$$\underline{R} = \{(a, b) \in \underline{A} \times \underline{B} \mid \underline{a \text{ is joined in } b}\}$$

Notice that $\underline{R \subseteq A \times B.}$

↳ Cross product

It is defined using the relation is joined in.

The set R is a binary relation from A to B .

- A binary relation R from a set A to a set B is a subset of $A \times B$.
- The domain of the relation consists of the first elements in R .
- The range consists of the second elements; they are denoted by $\text{dom}(R)$ and $\text{range}(R)$, respectively.

Note: A binary relation from A to itself is a binary relation on A .

Example: Let $A = \{2, 3, 5\}$ and $B = \{2, 3, 4, 6, 7\}$. Define a relation R from A to B as follows:

$$R = \{(a, b) \mid \underline{a \text{ is a factor of } b}\}$$

Definition is $R = \{(a, b) \mid (a, b) \in A \times B\}$, Given condition is a is a factor of b

$$R = \{(2, 2), (3, 3), (2, 4), (2, 6), (3, 6)\}$$

$\therefore \text{domain of } R = \{2, 3\}, \text{ Range} = \{2, 3, 4, 6\}$

Note

- If a set A has ' n ' elements, $A \times A$ has n^2 elements, $\therefore \boxed{2^{n^2}}$ relation from A to A .
- If A has ' m ' element and B has ' n ' element, then $A \times B$ has $m \times n$ elements $\therefore \underline{2^{m \times n}}$ relation from A to B .

Adjacency Matrix of the relation

Representation of Relations

Let $A = \{a_1, a_2, \dots, a_m\}$ and $B = \{b_1, b_2, \dots, b_n\}$

be finite sets containing m and n elements.

R is a relation from A to B . The relation R can be represented by an $m \times n$ matrix $M_R = [m_{ij}]$,

$$m_{ij} = \begin{cases} 1 & \text{if } a_i R b_j \\ 0 & \text{otherwise} \end{cases}$$

M_R is the adjacency matrix of the relation R .

Representation of Relations (Graphical Method)

A relation R on a finite set A can also be represented pictorially.

Each element of A is shown as a point, called a vertex (or node).

Every ordered pair $(a, b) \in R$ is shown as a directed arc/line segment, called an edge, from a to b .

The diagram formed is called a directed graph (or digraph).

If an edge (a, b) exists, then vertex b is said to be adjacent to vertex a .

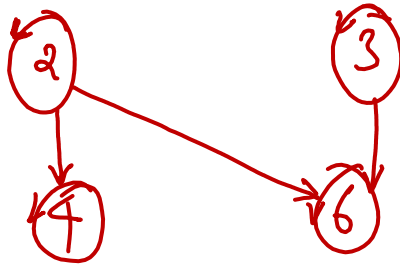
Represent the relation R defined on $A = \{2, 3, 4, 6\}$ by the phrase *is a factor of* in a digraph.

$$A = \{2, 3, 4, 6\}$$

define the relation $R = \{(a, b) \mid (a, b) \in A \times A, a \text{ is factor to } b\}$

$$R = \{(2, 2), (3, 3), (4, 4), (6, 6), (2, 4), (2, 6), (3, 6)\}$$

✎ Draw the digraph



Inverse Relation

Let R be any relation from a set A to a set B . The *inverse* of R , denoted by R^{-1} , is the relation from B to A .

$$R^{-1} = \{(b, a) \mid (a, b) \in R\}$$

$$R = \{(a, b) \mid (a, b) \in A \times B\}$$

Example: Consider the relation $\leq$ on the set $A = \{2, 3, 4\}$, find its inverse.

Soln

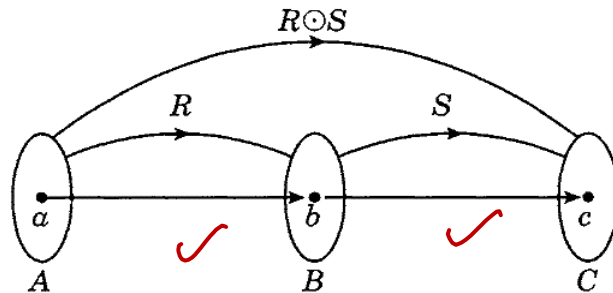
$$R = \{(2, 2), (3, 3), (4, 4), (2, 3), (2, 4), (3, 4)\}$$

$$R^{-1} = \{(2, 2), (3, 3), (4, 4), (3, 2), (4, 2), (4, 3)\}$$

Composition of Relations

- Let A , B , and C be sets, and let R be a relation from A to B , and let S be a relation from B to C .
- That is, R is a subset of $A \times B$ and S is a subset of $B \times C$. Then R and S give rise to a relation from A to C denoted by $R \circ S$.
- Defined by: $a(R \circ S)c$ if for some $b \in B$ we have aRb and bSc . That is
 $R \circ S = \{(a, c) \mid \text{there exists } b \in B \text{ for which } (a, b) \in R \text{ and } (b, c) \in S\}$

The relation $R \circ S$
is called the composition of R and S .



Let $A = \{1, 2, 3, 4\}$, $B = \{a, b, c, d\}$, $C = \{x, y, z\}$ and let

$R = \{(1, a), (2, d), (3, a), (3, b), (3, d)\}$ and $S = \{(b, x), (b, z), (c, y), (d, z)\}$.

Find the $(R \circ S)$.

$$A = \{1, 2, 3, 4\} \quad B = \{a, b, c, d\} \quad C = \{x, y, z\}$$

$$R = \{(1, a), (2, d), (3, a), (3, b), (3, d)\}$$

$$S = \{(b, x), (b, z), (c, y), (d, z)\}$$

$$R \circ S = \{(2, z), (3, z), (3, x)\}$$

